

Lecture 15

Perturbation Theory for Linear Operators

Recommended optional route: 7 recommended + 3 extensions

This sheet accompanies the optional, unassessed capstone. Work Exercises 15.1–15.7 for the recommended optional route (typically 60–75 minutes). Exercises 15.8–15.10 are extensions; allow roughly 100–125 minutes for all ten. Throughout, $H(\varepsilon) = H_0 + \varepsilon V$ with H_0 and V self-adjoint on a finite-dimensional complex inner-product space and $\varepsilon \in \mathbb{R}$, and denominators are written with the level of interest first.

In the infinite-dimensional extensions, $\mathcal{H}$ is complex, $\mathcal{B}(\mathcal{H})$ denotes bounded operators, and every unbounded operator is understood together with its stated domain.

Exercise 15.1 (computational) Let $H_0 = \text{diag}(1, 4)$ and $V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Find the first- and second-order shifts of both levels, and compare the second-order prediction with the exact eigenvalues at $\varepsilon = 0.1$.

Exercise 15.2 (computational) For $H_0 = \text{diag}(0, 2, 5)$ and

$$V = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 2 \\ 0 & 2 & 0 \end{pmatrix},$$

compute all three first- and second-order shifts, and verify the two checks $\sum_n E_n^{(1)} = \text{tr } V$ and $\sum_n E_n^{(2)} = 0$.

Exercise 15.3 (computational) For the three-level system of the lecture, $H_0 = \text{diag}(0, 1, 3)$ with V having rows $(2, 1, 1)$, $(1, 1, 1)$, $(1, 1, -3)$, compute the first-order correction to the *top* eigenvector and evaluate it at $\varepsilon = 0.1$.

Exercise 15.4 (computational) Let $H_0 = \text{diag}(0, 1, 3)$ and $V = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Compute $\|V\|$, and find the range of $|\varepsilon|$ for which the three *closed* disks of the spectral stability theorem are pairwise disjoint. State the supremal threshold.

Exercise 15.5 (proof) (*Hellmann–Feynman.*) Let $|n(\varepsilon)\rangle$ be a normalized eigenvector of $H(\varepsilon)$ depending differentiably on ε , with eigenvalue $E_n(\varepsilon)$. Prove that

$$\frac{dE_n}{d\varepsilon} = \langle n(\varepsilon) | V | n(\varepsilon) \rangle,$$

and deduce the first-order formula $E_n^{(1)} = \langle n | V | n \rangle$.

Exercise 15.6 (proof) Assume H_0 has simple eigenvalues. Prove that $\sum_n E_n^{(2)} = 0$, and that the second-order shift of the *lowest* level is always ≤ 0 while that of the highest is always ≥ 0 .

Exercise 15.7 (computational) Let $H_0 = \text{diag}(2, 2, 6)$ and

$$V = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Explain why the non-degenerate formulas cannot be used, then find the first-order branches of the degenerate level and their zeroth-order states. Finally explain why PVP does not guarantee splitting or a unique basis in general, using $H_0 = \text{diag}(1, 1, 4)$ and $V = I$.

Optional extensions/outlook: Exercises 15.8–15.10

Optional extension: complex branch points

Exercise 15.8 (★) For $\Delta > 0$ consider the two-level family

$$H(\varepsilon) = \begin{pmatrix} \Delta/2 & \varepsilon \\ \varepsilon & -\Delta/2 \end{pmatrix}.$$

(a) Find the exact eigenvalues. (b) Expand the upper one to order ε^2 and check it against second-order perturbation theory. (c) Regarding ε as a complex variable, find the radius of convergence of that expansion, and say what obstructs it.

Optional extension/outlook: infinite dimensions

Exercise 15.9 (computational) Let D be the diagonal operator $De_n = \frac{1}{n}e_n$ on ℓ^2 and let $z = -1$. Compute $\|(D - z \text{id})^{-1}\|$, and determine the range of $|\varepsilon|$ for which the resolvent expansion is *guaranteed uniformly using only the stated norm data*, together with its supremal radius, $(D + \varepsilon V - z \text{id})^{-1} = \sum_{k \geq 0} (-\varepsilon)^k R_0(z) (V R_0(z))^k$ for a bounded self-adjoint V with $\|V\| = 2$. Show that the uniform half-radius is sharp for $V = 2I$, whereas $V = 2P_1$ permits the larger range $|\varepsilon| < 1$, where P_1 projects onto $\text{span}(e_1)$.

Exercise 15.10 (★) The Bender–Wu analysis of the quartic anharmonic oscillator $H(\varepsilon) = \frac{1}{2}(\hat{p}^2 + \hat{x}^2) + \varepsilon \hat{x}^4$ finds divergent large-order behaviour and complex branch points accumulating at $\varepsilon = 0$. Explain why this gives zero convergence radius yet permits asymptotically useful low-order accuracy. Why is negative coupling only physical motivation, not a spectral proof without a chosen self-adjoint realization? Contrast the finite-dimensional simple-level case.